

Polycythemia / MPN — Approach to polycythemia (workup)



1. Erythrocytosis detected

WHAT YOU SEE

A red blood-filled test tube at top with a red arrow labelled ERYTHROCYTOSIS DETECTED — the entry sample that triggers the whole sorting machine.

WHAT IT ENCODES

Entry point: confirm a TRUE, sustained elevation before launching a workup. Diagnostic thresholds for erythrocytosis are Hgb >16.5 g/dL or Hct >49% in men, and Hgb >16.0 g/dL or Hct >48% in women (2016 WHO PV criteria). Always repeat the CBC — a single high value may be hemoconcentration. Mnemonic: 'TRUE before CHASE' — prove the red cell mass is really up before you order EPO, JAK2, and imaging.

BOARD TRAP

A WRONG move is to start phlebotomy or order a JAK2/EPO panel off one mildly elevated Hct. The discriminator: repeat the CBC and assess plasma volume first — transient dehydration or lab error mimics erythrocytosis, and treating it commits the patient to an unnecessary MPN workup.

2. Plasma volume / RCM

WHAT YOU SEE

GATE 1 plasma volume meter, CONTRACTED SMALL PLASMA BAG

WHAT IT ENCODES

Gate 1 separates TRUE polycythemia (absolute rise in red cell mass) from RELATIVE/spurious polycythemia (normal red cell mass with a contracted plasma volume). Hct is a ratio, so a shrunken plasma denominator falsely raises it. The historic gold standard is a chromium-51–labeled red cell mass study, but in practice very high Hgb/Hct (men Hgb >18.5, women >16.5) is taken as presumptive true erythrocytosis and RCM testing is reserved for borderline cases. Mnemonic: 'numerator vs denominator.'

BOARD TRAP

The trap is assuming every high Hct means more red cells. WRONG — if plasma volume is contracted (the small plasma bag), red cell mass can be normal. Discriminator: a normal calculated/measured RCM with low plasma volume = relative polycythemia, which needs hydration and med review, not a hematologic workup.

3. Relative (Gaisböck)

WHAT YOU SEE

A yellow disposal bucket far left labelled RELSÖCK'S / RELATIVE — the bin for spurious polycythemia, the first cause sorted out by Gate 1.

WHAT IT ENCODES

Relative (spurious/apparent) polycythemia = NORMAL red cell mass with reduced plasma volume. Causes: dehydration, diuretics, vomiting/burns, and Gaisböck syndrome (stress erythrocytosis in middle-aged hypertensive overweight male smokers). Management is treating the cause — rehydrate, stop diuretics, control BP, smoking cessation — not phlebotomy. Mnemonic: 'Gaisböck = the stressed, smoking, hypertensive businessman.'

BOARD TRAP

A classic boards stem: hypertensive overweight male smoker with mildly high Hct — the WRONG answer is polycythemia vera with phlebotomy. The discriminator is a NORMAL red cell mass and absence of splenomegaly/pruritus/JAK2; this is relative polycythemia, treated with lifestyle and addressing diuretics, not bloodletting.

4. Gate 1: need RCM

WHAT YOU SEE

Warning triangle Hgb<20 / Hct<60% → NEED RCM

WHAT IT ENCODES

When values are very high (e.g., Hct >60% in men, >56% in women), red cell mass is essentially always elevated and formal RCM measurement is unnecessary. In the intermediate zone (the warning-triangle Hgb <20 / Hct <60% range), an isotope red cell mass study can confirm true erythrocytosis when the diagnosis is unclear. Mnemonic: 'sky-high Hct skips the RCM step.'

BOARD TRAP

WRONG to demand a chromium-51 RCM study in a patient with Hct 65% before acting — at extreme values RCM is presumed elevated. Discriminator: reserve formal RCM for borderline counts; chasing it at extreme values delays diagnosis and the isotope test is rarely available.

5. JAK2 detector >95%

WHAT YOU SEE

GATE 2 JAK2 DETECTOR >95% V617F, EXON 12

WHAT IT ENCODES

Gate 2 molecular testing: JAK2 V617F (exon 14) is present in ~95% of polycythemia vera; the remaining ~3–4% carry JAK2 exon 12 mutations, so a V617F-negative but clinically suspicious case gets reflex exon 12 testing. JAK2 is a constitutively active tyrosine kinase driving EPO-independent erythroid proliferation. A positive JAK2 is a MAJOR WHO criterion for PV. Mnemonic: 'JAK2 — the master switch jammed ON.'

BOARD TRAP

The trap: a positive JAK2 V617F is NOT specific for PV — it also appears in ~50–60% of ET and ~50% of primary myelofibrosis. So JAK2+ alone does not equal PV. Discriminator: combine JAK2 with a SUPPRESSED EPO and red-cell predominance; JAK2+ with high EPO and isolated thrombocytosis points elsewhere.

6. EPO level meter

WHAT YOU SEE

EPO LEVEL METER gauge

WHAT IT ENCODES

Serum EPO is the pivotal discriminating test. In PV the autonomous JAK2-driven erythropoiesis suppresses EPO via negative feedback, so EPO is LOW (or low-normal). In secondary polycythemia the rise is EPO-DRIVEN, so EPO is HIGH or inappropriately normal. A low EPO is itself a MINOR WHO criterion for PV. Mnemonic: 'LOW EPO = Lesion in the marrow (PV); HIGH EPO = Hypoxia/tumor (secondary).'

BOARD TRAP

High EPO does NOT mean PV. WRONG answer — students invert the logic. In PV the marrow makes red cells on its own and feedback SUPPRESSES EPO (low). High/normal EPO points to a SECONDARY cause (hypoxia, tumor, doping). Discriminator: low EPO + JAK2+ = PV; high EPO = go hunting for hypoxia or an EPO source.

7. Low EPO → PV

WHAT YOU SEE

A red bucket on the far right labelled PV — the final correct bin reached only after low EPO + JAK2; the destination of the true-polycythemia pathway.

WHAT IT ENCODES

Suppressed EPO + JAK2 mutation = polycythemia vera. WHO 2016 PV diagnosis = all 3 major OR first 2 major + the minor criterion: (1) elevated Hgb/Hct or RCM, (2) bone marrow trilineage panmyelosis with pleomorphic megakaryocytes, (3) JAK2 mutation; minor = subnormal EPO. Treat with phlebotomy to Hct <45% plus low-dose aspirin; add hydroxyurea (or ruxolitinib/interferon) if high-risk (age >60 or prior thrombosis). Mnemonic: 'PV = Pruritus, Plethora, Phlebotomy.'

BOARD TRAP

WRONG to call any erythrocytosis with low EPO 'PV' without confirming JAK2 — congenital high-O₂-affinity hemoglobinopathies and EPO-receptor mutations can also have low EPO. Discriminator: PV needs the JAK2 mutation (or exon 12); JAK2-negative low-EPO erythrocytosis in a young patient suggests a congenital cause.

8. Secondary / exclude PV

WHAT YOU SEE

SECONDARY, EXCLUDE PV label on EPO meter

WHAT IT ENCODES

An elevated or inappropriately normal EPO shifts the diagnosis to SECONDARY polycythemia and effectively excludes PV (where EPO should be suppressed). Next steps: pulse oximetry/ABG and carboxyhemoglobin for hypoxia, then imaging (renal/hepatic/CNS) to hunt for an EPO-secreting tumor if no hypoxia is found. Mnemonic: 'EPO up → look UP the airway and INSIDE the organs.'

BOARD TRAP

The trap is treating secondary polycythemia like PV with aggressive phlebotomy. WRONG — in hypoxia-driven secondary polycythemia the high Hct is compensatory; aggressive phlebotomy can worsen tissue oxygen delivery. Discriminator: fix the underlying cause (oxygen, CPAP, resect tumor); phlebotomy is reserved for symptomatic hyperviscosity.

9. Hypoxia

WHAT YOU SEE

A blue bucket labelled HYPOXIA — the secondary-cause bin for high-EPO patients whose driver is low tissue oxygen.

WHAT IT ENCODES

Most common cause of secondary polycythemia: chronic tissue hypoxia driving appropriate EPO release. Causes: COPD, obstructive sleep apnea (Pickwickian/obesity-hypoventilation), high-altitude residence, cyanotic congenital heart disease with right-to-left shunt, and smoking (carboxyhemoglobin causes a left-shift and functional hypoxia). Check SaO₂ and carboxyhemoglobin. Mnemonic: 'high CO from cigarettes hides as normal SaO₂.'

BOARD TRAP

A boards trap: a heavy smoker with high Hct and a NORMAL pulse-ox SaO₂ — calling it PV is WRONG. Pulse oximetry cannot detect carboxyhemoglobin; the discriminator is an elevated carboxyhemoglobin level (co-oximetry), revealing smoker's polycythemia, which resolves with cessation.

10. EPO-secreting tumors

WHAT YOU SEE

A dark-red bucket labelled EPO TUMORS — the secondary-cause bin for autonomous EPO-secreting tumors.

WHAT IT ENCODES

Tumors that autonomously secrete EPO cause secondary polycythemia with HIGH EPO and no hypoxia. Classic four: Renal cell carcinoma, Hepatocellular carcinoma, cerebellar Hemangioblastoma, and uterine leiomyomata (fibroids); also pheochromocytoma and parathyroid adenomas. Mnemonic: 'EPO tumors — Renal, Hepatic, Hemangioblastoma, uterine fibroids (the 4 H-driven sources).'

BOARD TRAP

WRONG to stop at 'secondary polycythemia' once EPO is high without imaging for a tumor. Discriminator: a non-smoker with high EPO, no hypoxia, and a flank mass or new headache needs CT abdomen/MRI brain — missing an EPO-secreting RCC or cerebellar hemangioblastoma (think von Hippel–Lindau) is the tested error.

11. Renal causes

WHAT YOU SEE

An orange bucket labelled RENAL — the secondary-cause bin for kidney-driven EPO elevation.

WHAT IT ENCODES

Renal pathology raises EPO either by local hypoxia or autonomous production: post-renal-transplant erythrocytosis (10–15% of transplants, treat with ACE inhibitor/ARB), renal artery stenosis, polycystic kidney disease, hydronephrosis, and renal cell carcinoma. Mnemonic: 'a kidney that's squeezed, cystic, or cancerous cranks out EPO.'

BOARD TRAP

Post-transplant erythrocytosis treated with phlebotomy alone is the WRONG approach. Discriminator: first-line therapy is an ACE inhibitor or ARB, which lowers EPO and Hct; reserving phlebotomy for refractory/symptomatic cases is the tested point.

12. Drugs (testosterone)

WHAT YOU SEE

A green bucket labelled DRUGS — the secondary-cause bin for testosterone/androgen- and EPO-doping-driven erythrocytosis.

WHAT IT ENCODES

Drug/iatrogenic erythrocytosis: exogenous testosterone and anabolic-androgenic steroids stimulate erythropoiesis (a dose-limiting side effect of testosterone replacement — monitor Hct and hold if >54%); recombinant EPO/darbepoetin doping in athletes; and SGLT2 inhibitors mildly raise Hct. Mnemonic: 'the gym/anti-aging clinic patient with a high Hct = testosterone.'

BOARD TRAP

An older man on testosterone replacement with Hct 56% being diagnosed as PV is the WRONG call. Discriminator: medication history reveals exogenous androgen; management is dose reduction/holding testosterone and therapeutic phlebotomy — not committing to lifelong MPN therapy. EPO is typically LOW here too (androgens suppress endogenous EPO), so don't be fooled into the PV pathway.

13. Familial / high-affinity Hb

WHAT YOU SEE

A tan bucket labelled FAMILIAL — the secondary-cause bin for congenital high-oxygen-affinity hemoglobin and other inherited erythrocytoses.

WHAT IT ENCODES

Congenital erythrocytosis: high-oxygen-affinity hemoglobin variants (left-shifted O₂ dissociation curve, low p₅₀ — Hb releases O₂ poorly so tissues sense hypoxia), EPO-receptor truncation mutations (primary familial polycythemia, LOW EPO), 2,3-BPG deficiency, and VHL Chuvash polycythemia. Suspect in young patients with a family history. Mnemonic: 'young + family history + low p₅₀ = high-affinity Hb.'

BOARD TRAP

A young adult with lifelong high Hct and JAK2-negative testing labeled 'PV' is WRONG. Discriminator: order a hemoglobin-oxygen dissociation curve (p₅₀) and family studies — a low p₅₀ indicates a high-affinity hemoglobin, a benign congenital condition, not a clonal MPN.

14. Arterial O₂ saturation probe

WHAT YOU SEE

An ARTERIAL O₂ SATURATION PROBE clipped onto the EPO/secondary line — the next test after a high EPO, measuring oxygenation to find a hypoxic cause.

WHAT IT ENCODES

After EPO is found high/normal, measure arterial oxygenation: SaO₂ by ABG (not just pulse ox) and carboxyhemoglobin by co-oximetry to detect smoker's/CO-mediated hypoxia. SaO₂ <92% supports a hypoxic secondary cause. If oxygenation is normal, pivot to imaging for an EPO-secreting tumor. Mnemonic: 'EPO high → check the O₂ probe → if normal, hunt the tumor.'

BOARD TRAP

Relying on a normal finger pulse-ox to exclude hypoxia is the WRONG move. Discriminator: pulse oximetry misses carboxyhemoglobin (smokers) and overnight desaturation (OSA); the tested next steps are co-oximetry/ABG and, when suspected, a sleep study before concluding the polycythemia is unexplained.
